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A fast and efficient way of obtaining the average molecular weight of block copolymers *via* DOSY†

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Diffusion-ordered NMR spectroscopy (DOSY) allows for solvent-independent molecular weight calibration with low lab variability. Methyl methacrylate standards have been used to calibrate DOSY data for various solvents. Following the Stokes–Einstein equation, a high correlation of the data was obtained when the viscosity of the solvent was taken into account. Moreover, the inter-laboratory and inter-polymer consistency was confirmed by using the pMMA calibration obtained on an NMR system in Germany, which was applied to polystyrene diffusion coefficients that were obtained in Australia. Furthermore, the molecular weight of polymers with added complexity, such as block copolymers, is determined with diffusion data. Butyl acrylate (BA), methyl acrylate (MA), PEG methyl ether acrylate (PEGMEA) and MMA are used as building blocks to synthesize the block copolymers. Firstly, DOSY calibration curves are prepared for the synthesized homopolymers pBA and pPEGMEA. Then, the diffusion data are compared to block copolymers with different compositions of pBA-*b*-pMA, pBA-*b*-pPEGMEA and pMMA-*b*-pPEGMEA and an accurate determination of the molecular weight of the block copolymers could be achieved using the homopolymer calibration curves.

Introduction

The determination of molecular weight is an essential analytical procedure in polymer chemistry.[1] Various sophisticated techniques have been developed to determine the molecular weight of polymers, including size exclusion chromatography (SEC), light scattering, or osmometry.[2,3] However, these methods often have limitations such as low sensitivity and a limited range of accessible molecular weights.[4–6] Diffusion-ordered spectroscopy (DOSY) has emerged as a promising technique for determining the molecular weight of polymers.[7,8] Based on many previous studies, we have demonstrated that DOSY is an accurate method for assessing the average molecular weights of polymers, competing favourably with standard SEC. Furthermore, we have shown that even if standard calibration is applied to a very different type of polymer, a reasonable molecular weight estimate is obtained with an accuracy that at the very least rivals that of standard universal calibration SEC.[7] Most importantly, however, we have demonstrated that DOSY calibration is truly universal in the sense that no individual calibration of instruments is required. One calibration obtained in one lab is in principle applicable worldwide in a very broad molecular weight range from oligomers to millions of Da. This removes the need for constant re-calibration of instruments and makes DOSY much more accessible as a molecular weight determination tool compared to SEC, the current laboratory gold standard. While still more interlaboratory studies will be required to establish DOSY as a fully robust method, the results presented in the literature to date are very promising. A detailed description of DOSY and a practical guide to performing DOSY for molecular weight determination are given in our previous work.[7] Yet, so far, we only discussed the application of DOSY to linear homopolymers. Due to the rise of techniques such as controlled polymerization methods, polymer chemists have nowadays gained access to complex polymers in an abundant and easily accessible way.[9] Furthermore, macromolecules with precise tacticity, monomer sequences and topologies, such as nucleic acids, proteins and polysaccharides, are found throughout nature, which gives rise to materials with specific and exciting characteristics that require expert characterization.[10,11] Hence,

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for the next step with regard to DOSY molecular weight analysis, we focus in this work on block copolymers to cover one of the most common types of complex polymers.[12]

Block copolymers are polymeric materials consisting of two or more chemically distinct blocks that are covalently linked. These materials have gained significant interest from the scientific community due to their unique combination of mechanical, thermal, and optical properties, making them suitable for a wide range of applications in various industries.[13] The importance of block copolymers stems from their ability to self-assemble into complex nanostructures, which can be precisely controlled by choice of block chemistry, molecular weight, and processing conditions.[14] In a solution-based self-assembly process, the core of the micelle is formed by insoluble or solvophobic blocks. The soluble or solvophilic block forms the corona or the shell of the micelle. This combination leads to the colloidal stabilization of the micelle in the chosen solvent.[15] As useful as these properties are, such block polymers can for the same reason also be challenging to analyze. In fact, using standard SEC based on universal calibration has already proven to be very difficult.[16] Moreover, the limitation of choosing the solvent on which the SEC is operated further limits the determination of the size of the block copolymer.

To demonstrate that DOSY is also an accurate way to determine the average molecular weight of block polymers, we first extend our method by providing calibration data for some additional homopolymers. Then, we analyze diblock copolymers, for which homopolymer calibrations are available, and compare the outcome of these calibrations with the block copolymers. In Fig. 1, the general principle of this study is described. When a homopolymer standard DOSY calibration is used (orange), can a block copolymer (bicolor) still be determined on the same calibration within reasonable error margins? Indeed, a reasonable molecular weight estimate is still obtained as we will demonstrate below. We performed these studies on acrylate and methacrylate polymers and mixed block copolymers to stress-test the method. A specific block copolymer of interest here was synthesized from methyl methacrylate and polyethylene glycol methyl ether acrylate. The resulting block copolymers are interesting because they have the tendency to self-assemble and feature highly disparate solubility and significantly different chain stiffness in their individual blocks.[17] Additionally, the PEG side-groups of acrylates result in a brush-like (dispersed) polymer structure, which poses another challenge for classical analysis.[18]

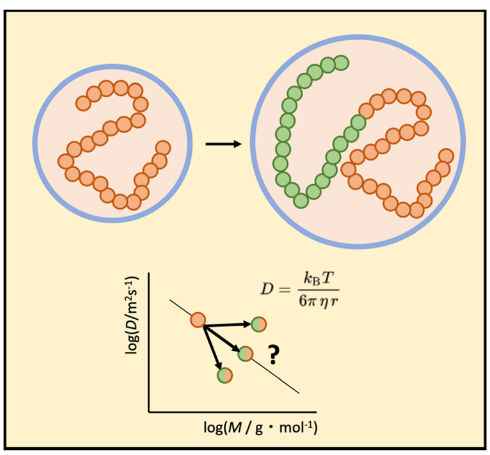

Fig. 1 The research question of this work: if a block copolymer is formed, can the residual polymer be measured in DOSY on the same molecular weight calibration as the starting homopolymer, or not?

SEC requires the addition of sophisticated detectors such as multi-angle laser light scattering (MALLS) to determine the branching structure and size of the branches in the copolymer. Should the polymers described above prove to yield reasonable results with simple DOSY calibrations, many daily issues of polymer chemists could be solved through a simple and practical approach by means of a single DOSY experiment.

Results and discussion

Poly(methyl methacrylate) homopolymer calibration

In our previous work, we reported calibrations for polystyrene and poly(ethylene glycol) standards. Other researchers provided DOSY calibrations for a number of other polymers, yet poly(methyl methacrylate), pMMA, seemed to be still worthwhile to study closely. Hence, we performed full DOSY calibrations on a series of pMMA standards in a variety of typical deuterated NMR solvents. Generally, the diffusion coefficient D of a (by approximation) spherical object, such as a polymer coil, correlates with its hydrodynamic radius r_h *via* the Stokes–Einstein relation, where T is the absolute temperature, k_B is the Boltzmann constant and η is the solvent bulk viscosity:

$$D = \frac{k_B T}{6\pi\eta r_h} \quad (1)$$

r_h can be correlated to the average molecular weight of the polymer *via* the empirical Rouse–Zimm model, resulting in eqn (2).

$$D = b' M^{-\nu} \quad (2)$$

The logarithm of the obtained diffusion coefficient is plotted against the logarithm of the molecular weight to obtain a direct calibration in any given solvent. According to eqn (2), a linear plot should be obtained. The collected data for diffusion coefficients in a variety of deuterated solvents are shown in Fig. 2. Indeed, also for pMMA, clear linear correlations apply to the whole tested mass range of 7900 to 106 000 g mol^{-1}. The top part of Fig. 2 depicts the individual calibrations per solvent. In the lower part, we plotted the logarithm of $D \times \eta$. As we have shown before,[7] this provides for a universal solvent-independent calibration and all calibration curves overlap almost perfectly.

The results of the individual fits are given in Table 1. Compared to recent data reported by other groups, a good match was found with the DOSY calibration obtained by Arrabal-Campos with a divergence of less than 15%[16] (see Fig. S1†). The difference between the present work and their

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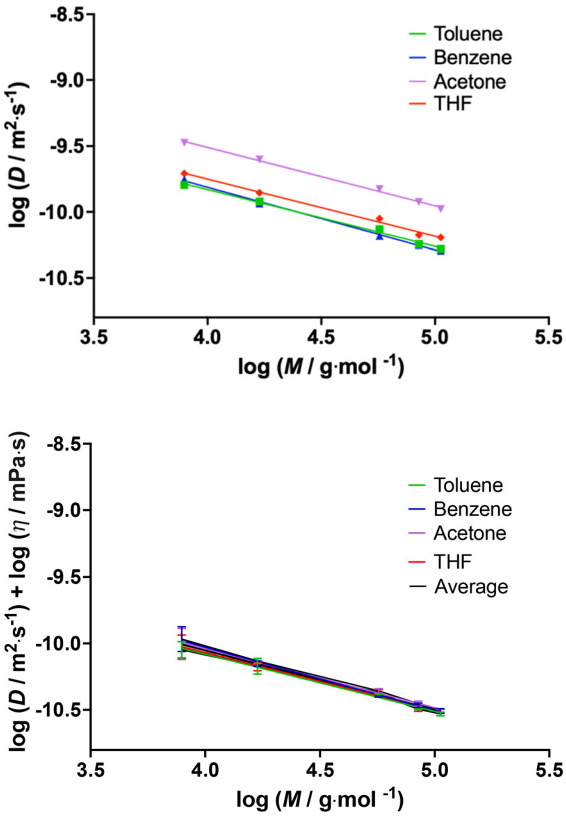

Fig. 2 Diffusion coefficients determined *via* DOSY for pMMA standards in a series of solvents (top) and viscosity-corrected data for the same measurements (bottom).

Table 1 Calibration curves for pMMA in different solvents. The last line refers to a fit of log ($D \times \eta$) *vs.* log(M)

Sample	$\lg(b'/\mathrm{m^2\ s^{-1}})$	ν
Toluene	−8.11	0.43
Benzene	−7.91	0.48
Acetone	−7.74	0.44
THF	−8.02	0.43
$D \times \eta$ corrected	−8.27	0.45

study is the highest for the high molecular weight polymers. This may be due to differences in the solvent affinity and hence the coil size of the polymers in each medium, as well as the variation in the processing parameters, as the experimental error generally increases with the molecular weight. Compared to the data reported by Ruzicka *et al.*,[19] an error with a maximum deviation of 24% was found over the same range in which our pMMA standards were measured. This difference is still in an acceptable range, knowing that SEC is equally associated with an error of 10–20%. Some more refinement might, however, be required in the future to establish a better instrument-independent calibration, as different approaches in the measurement might still cause discrepancies between labs. At this stage, it is not fully clear what causes occasional discrepancies between laboratories, yet this and other interlaboratory exercises provide enough evidence for the hypothesis that the DOSY calibrations are independent of the field strength or the NMR manufacturer. It is more likely that differences in detail in the pulse sequence are responsible, as much as the temperature for which the calibration is provided.

With these data, we further tested our hypothesis of inter-laboratory/inter-polymer consistency. The pMMA calibration was obtained on an NMR system in Germany and we applied this calibration to diffusion coefficients obtained in Australia for a polystyrene sample with an M_n value of 18 000 g mol^{-1}. The diffusion coefficient of the said sample is 1.32 10^{-10} m^2 s^{-1}. The THF pMMA calibration (Table 1) yields an estimate of 19 840 g mol^{-1}. This represents an error of a mere 10% and is hence equal to or better than the estimation obtained using a directly calibrated SEC system.[20] This confirms the use of a DOSY system to predict the molecular weight, independent of where a calibration was obtained and at the same time underpins that the actual type of polymer is not as important in DOSY compared to SEC.

Block copolymer pBA-pMA

Next, we turned to an all-acrylate block copolymer, namely p(butyl acrylate)-*b*-p(methyl acrylate), pBA-pMA. Homopolymer calibrations of poly(methyl acrylate), pMA, are available in the literature ($\lg(b'/\mathrm{m^2\ s^{-1}}) = -7.72$, $\nu = 0.5$, measured in toluene).[21,22] For the second block, poly(butyl acrylate), pBA, no DOSY calibration is available so far. Hence, we measured the diffusion coefficients of RAFT-synthesized pBA homopolymers *via* DOSY and calibrated against weight average molecular weight results obtained using SEC-MALLS for the same samples. Absolute molecular weights are given in Table 2 in comparison with the results typically obtained from universal calibration SEC employing MHKS values for pBA. As one can see, conventionally calibrated SEC does not perform overly well in comparison to light scattering detection. The RAFT polymerization was well controlled achieving a low dispersity (see the conventional SEC results in Table 2). Different degrees of polymerization were targeted in the range of 30 to 100. In line with all other polymers tested, the diffusion coefficients from DOSY exhibit a linear relation with the weight average molecular weight of the pBA homopolymers, obtained *via* SEC-MALLS. The DOSY calibration shows excellent linearity and an evenly low error in the entire covered molecular weight

Table 2 MALLS and SEC results of poly(butyl acrylate), pBA, and samples with various DPs

Sample	MALLS M_w (g mol^{-1})	SEC M_w[a] (g mol^{-1})	SEC[a] Đ
pBA35	4500	6800	1.07
pBA55	6100	7800	1.14
pBA60	5400	7200	1.12
pBA75	9700	12 200	1.11
pBA80	7400	8600	1.09
pBA100	11 800	10 600	1.19

[a] SEC molecular weight was determined on a conventional SEC calibrated on PS standards, using MHKS parameters for pBA.

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range. The individual diffusion coefficients and information on the RAFT polymerizations are found in the ESI.† The calibration fit results for various solvents are given in Table 3 (and plotted in Fig. S7†). Part of the pBA samples used for calibration were successively chain extended with pMA. This created a library of different block copolymers that can be comprehensively tested for their diffusion coefficients and DOSY calibration behaviour. The results for this series are given in Table 4 and Fig. 3. It should be noted that here (and all other instances in this work), MALLS refers to weight-average molecular weight M_w, while for DOSY it is not 100% clear what average is measured and hence the abbreviation MW is chosen (assuming though that it is likely to be closer to the weight average).

The diffusion coefficients of the parent pBA homopolymers and their pMA chain extensions are plotted in Fig. 3 as a function of their molecular weight as determined by MALLS. It is clearly evident that all fit very well onto a single linear fit, indicating that they follow the same molecular weight calibration, irrespective of whether they are on a homo or block copolymer calibration. Due to the variation in conversions of different polymerizations, the homopolymer pBA with a targeted degree of polymerization (DP) of 55 and the block copolymer pBA-*b*-pMA with a targeted DP of 35 for the first block and 60 for the polymer extension have similar molecular weights and comparable diffusion coefficients. This again shows the power of this technique in which calibration curves can be used interchangeably.

Table 3 Calibration parameters for pBA in various solvents

Sample	$\lg(b'/\mathrm{m^2\ s^{-1}})$	ν
Benzene	−7.83	0.51
Acetone	−7.56	0.50
Chloroform	−7.86	0.50
$D \times \eta$ corrected	−8.05	0.51

Table 4 MALLS, SEC and DOSY results of pBA-pMA block copolymer samples with various DPs. M_w obtained *via* DOSY is obtained *via* the butyl acrylate calibration

Sample	MALLS M_w (g mol^{-1})	SEC[a] M_w (g mol^{-1})	SEC[a] $Đ$	DOSY MW (g mol^{-1})
pBA35-pMA30	6100	7700	1.15	6060
pBA35-pMA60	8000	10 200	1.22	7710
pBA35-pMA90	8900	9600	1.10	8350
pBA35-pMA120	9500	11 100	1.16	8690
pBA55-pMA30	7600	9200	1.16	7330
pBA55-pMA60	8700	9500	1.16	8460
pBA55-pMA90	9400	11 000	1.12	8930
pBA55-pMA120	10 200	11 100	1.16	9450
pBA75-pMA60	11 200	12 500	1.15	10 960
pBA75-pMA90	13 800	16 100	1.15	13 310
pBA75-pMA120	14 200	14 700	1.12	13 780

[a] SEC molecular weight was determined on a conventional SEC calibrated on PS standards, no MHKS parameters were used.

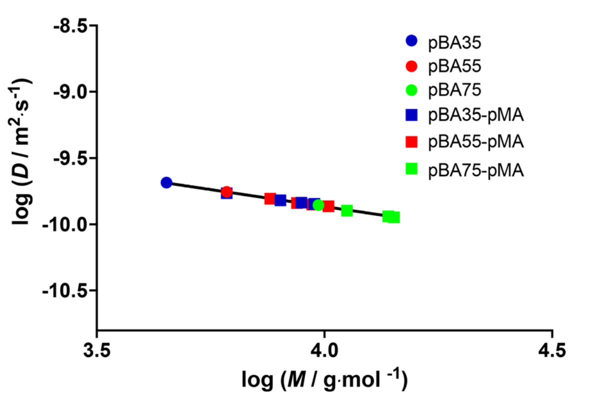

Fig. 3 DOSY molecular weight calibration for RAFT-made pBA and pBA-pMA polymers in deuterated benzene. Molecular weights were obtained *via* MALLS. The black line represents the best linear fit of all data, yielding the relation $\log(D) = -0.5135\ \log(M) - 7.812$.

pMMA-pPEGMEA block copolymer calibration

pBA and pMA are relatively similar polymers and indeed, their individual calibrations are very close to each other; hence a joint calibration with a close fit between the homo and block copolymers is partly expected. Therefore, we tested a block copolymer composed of pMMA and poly(polyethylene glycol methyl ether acrylate), pPEGMEA, which is a significantly more complex case. The molecular weight of PEGMEA is 480 g mol^{-1}. These polymers are much more complex due to being the mixture of acrylate and methacrylate blocks, containing pendant PEG side chains. Furthermore, these block copolymers are amphiphilic and are regularly used for biomedical application. In contrast to the pMA-*b*-pBA system, the blocks have very different solubility. PEGMEA is easily polymerized *via* RAFT. A reasonable progression in the molecular weight and comparatively low dispersities (for a brush polymer) were obtained (see Table 5), and the polymers obtained for a higher

Table 5 MALLS and SEC results of PEGMEA samples with various DPs

Sample	MALLS M_w (g mol^{-1})	SEC M_w[a] (g mol^{-1})	SEC $Đ$
pPEGMEA15	6500	7800	1.33
pPEGMEA20	10 500	7800	1.23
pPEGMEA40	16 200	10 100	1.36
pPEGMEA60	19 700	10 900	1.37
pPEGMEA80	19 600	11 100	1.35
pPEGMEA100	19 800	11 600	1.38

[a] SEC molecular weight was determined on a conventional SEC calibrated on PS standards, using MHKS parameters for PEG.

Table 6 DOSY molecular weight calibration parameters for pPEGMEA in different solvents

Sample	$\lg(b'/\mathrm{m^2\ s^{-1}})$	ν
Water	−8.31	0.47
Acetone	−8.01	0.39
$D \times \eta$ corrected	−8.39	0.43

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Table 7 MALLS and DOSY results of pMMA-pPEGMEA block copolymer samples with various degrees of polymerization

Sample	MALLS. M_w (g mol^{-1})	DOSY MW (g mol^{-1}) (pMMA)	Error (%)	DOSY MW (g mol^{-1}) (pPEGMEA)	Error (%)	DOSY MW (g mol^{-1}) (pMMA-pPEGMEA)	Error (%)
pMMA25	2630	2740	4	1540	71	2160	22
pMMA45	4910	4970	1	3010	63	3950	24
pMMA65	7010	7400	5	4710	49	5910	19
pMMA25-pPEGMEA25	10 600	14 690	28	10 200	4	11 810	10
pMMA25-pPEGMEA50	23 000	32 600	29	25 100	8	26 430	13
pMMA25-pPEGMEA75	45 400	56 160	19	46 340	2	45 780	1
pMMA45-pPEGMEA25	18 000	22 580	20	16 580	9	18 230	1
pMMA45-pPEGMEA50	34 500	43 860	21	35 070	2	35 670	3
pMMA45-pPEGMEA75	68 400	78 110	12	67 250	2	63 900	7
pMMA65-pPEGMEA25	31 500	41 540	24	32 980	4	33 760	7
pMMA65-pPEGMEA50	91 100	90 450	1	79 350	15	84 100	8
pMMA65-pPEGMEA75	135 000	139 730	3	129 610	4	115 000	18

target DP also showed lower conversions, and hence yielded quite similar molecular weights.

Again, before studying the block copolymers *via* DOSY, we first obtained a DOSY calibration for the homopolymer pPEGMEA. All samples fall in line, and a clear calibration can be obtained; see Fig. S2.† Table 6 summarizes the calibration results for water and acetone, and the solvent-independent calibration.

As mentioned, for pMMA-*b*-pPEGMEA polymers it is more challenging to obtain a single calibration, even if the individual calibration parameters look similar. Table 7 summarizes the molecular weights obtained from MALLS and DOSY. The diffusion coefficients were measured in deuterated acetone, which is a suitable solvent for both blocks. The same polymers were also measured in deuterated chloroform, which qualitatively showed the same results (see Table S4†). For the DOSY data, three different calibrations were applied, that is the two individual homopolymer calibrations discussed above, and a joint calibration that is constructed from all data being fitted jointly, as given in Fig. 4.

The error analysis, being the deviation between the DOSY result and the original MALLS measurement, reveals interesting insights. A pure pMMA calibration yields an overall error-prone result for the block copolymers, averaging at around 18%, but going up as high as almost 30% for individual measurements. By applying the mixed calibration as given in Fig. 4, much better results are obtained, with an average error of only 9%, when only the block copolymers are considered. Most interestingly though, the best results are obtained when a pure pPEGMEA calibration is applied: in this case, the average error amounts to only 6%.

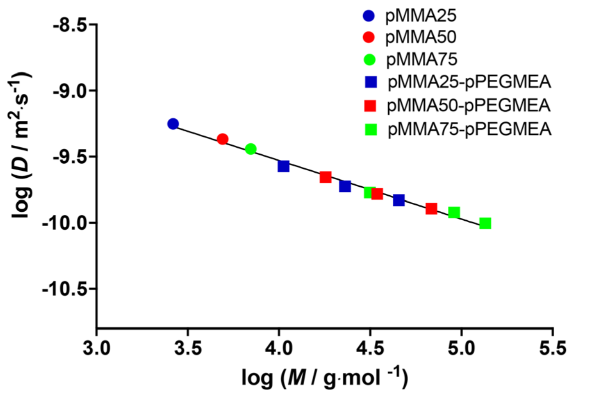

Fig. 4 Mixed DOSY calibration of pMMA homopolymers and pMMA-pPEGMEA block copolymers in deuterated acetone. The black line represents the best fit with all data combined with $\log(D) = -0.4355 \log(M) - 7.797$.

Table 8 MALLS and DOSY results of pBA-pPEGMEA block copolymer samples with various degrees of polymerization

Sample	*MALLS* M_w (g mol^{-1})	DOSY *MW* (g mol^{-1}) (pBA)	Error (%)	DOSY *MW* (g mol^{-1}) (pPEGMEA)	Error (%)	DOSY *Mw* (g mol^{-1}) (pBA-pPEGMEA)	Error (%)
pBA60	5400	5410	1	4020	34	4930	19
pBA80	7400	7370	1	5930	25	6900	14
pBA100	11 800	11 820	1	10 770	10	11 550	7
pBA60-pPEGMEA25	13 600	14 190	4	13 560	1	14 100	4
pBA60-pPEGMEA50	22 700	22 900	1	24 820	9	23 760	4
pBA60-pPEGMEA75	30 900	29 660	4	34 410	10	31 500	9
pBA80-pPEGMEA25	14 700	14 310	3	13 700	7	14 230	4
pBA80-pPEGMEA50	24 300	21 230	15	22 560	8	21 890	3
pBA80-pPEGMEA75	33 400	25 960	29	29 070	15	27 240	7
pBA100-pPEGMEA25	15 100	14 070	7	13 410	13	13 960	4
pBA100-pPEGMEA50	25 400	22 410	13	24 150	5	23 200	4
pBA100-pPEGMEA75	35 500	27 890	27	31 830	12	29 450	8

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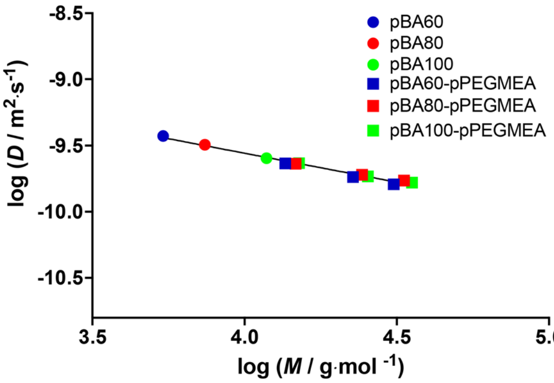

Fig. 5 Mixed DOSY calibration of pBA homopolymers and pBA-pPEGMEA block copolymers in deuterated acetone. The black line represents the best fit with all data combined with $\log(D) = -0.4542\log(M) - 7.750$.

BA-PEGMEA block copolymer calibration

To see if the same trend also holds for other block copolymers, we further carried out DOSY measurements on pBA-*b*-pPEGMEA polymers. Table 8 and Fig. 5 summarize the respective results in the same order as for pMMA-*b*-pPEGMEA polymers. In this case, the individual homopolymer calibrations show a larger discrepancy, which is directly evidenced when examining the data shown in Table 7. With increasing content of the second monomer, the homopolymer calibration of pBA fails. While some measurements show errors of only 3–4%, the overall average is about 11%, and as high as 30% for individual measurements. However, for the block copolymer, the mixed calibration (see Fig. S6†) yields the best result, with an overall low average error of 5%. The pure pPEGMEA calibration yields an average error of 9%.

Conclusions

The high quality of viscosity-corrected calibrations of linear pMMA, pBA and pPEGMEA further proves that DOSY is a very capable technique for determining the molecular weights of polymers. It is satisfying that pPEGMEA, which constitutes a brush polymer, still shows a very reasonable calibration. Furthermore, we demonstrated that calibrations are applicable across laboratories and NMR machines. Hence, the individual calibrations provided here are applicable to any lab worldwide, at least as long as the same methodology is used to obtain DOSY diffusion coefficients. However, even if comparatively low, our pMMA calibration deviates somewhat from those found in the literature, indicating that there are still outstanding issues in standardizing DOSY measurements. More experiments with respect to this are currently ongoing in our laboratory. Also, the establishment of SEC as a reliable molecular weight determination technique took several years if not decades to fully mature and to provide accurate measurements. In any case, the inter-laboratory errors already at this point in DOSY seem to be very acceptable when compared to SEC. A further point of attention is the measurement of polymers with distinct dispersity. The standard DOSY experiment as described herein does not allow the determination of the dispersity of a polymer, which is equally a question of further research. However, for calibration purposes, it is the best way forward to employ narrowly dispersed polymers to create an exact relation between the MW and the diffusion coefficient. These calibrations should then later also hold when procedures for molecular weight distribution analysis are introduced, analogous to the classical SEC.

The more significant outcome of this study is that DOSY is also well applicable to block copolymers. We found almost perfect DOSY calibration for pMA-*b*-pBA BCPs. For the mixed BCP pMMA-*b*-pPEGMEA, we found that either a pPEGMEA or a mixed BCP calibration yielded the best results, hence allowing the use of the standard pPEGMEA calibration to obtain molecular weights for their descendant block copolymers with reasonable accuracy (6%). For pBA-*b*-pPEGMEA, the mixed calibration performed slightly better, yet also here, the pPEGMEA homopolymer calibration resulted in reasonable accuracies of 9%. In both cases, however, it seemed better to use the pPEGMEA calibration rather than the calibration of the homopolymer of the first block. This is reasonable given the vast difference between linear (meth)acrylate and the brush polymer pPEGMEA, which likely dominates the overall physical properties of the BCP.

While it is difficult to derive a strict rule from these investigations, it seems to be recommendable to use the homopolymer DOSY calibration of the block with the larger overall mass in BCP analysis. Error limits in almost all cases are well within the expected range for SEC analysis, or even better. This is particularly true for a polymer such as pMMA-*b*-pPEGMEA, which due to its highly amphiphilic nature is notoriously difficult to assess in classical SEC.

Regardless, in this work, further indication is given that the universal calibration nature of DOSY does not only apply to the use of different solvents but also stretches to many other types of polymers and that a general calibration for all linear polymers could be within reach. This is in contrast to SEC, where the polymer studied herein would show quite high deviations (as also evidenced by the conventional SEC data given in the various tables here) if a universal calibration with or without correction *via* Mark–Houwink parameters is applied.

Author contributions

Pieter-Jan Voorter: conceptualization, methodology, visualization, investigation, validation, and writing – original draft. Manfred Wagner: conceptualization and supervision. Christine Rosenauer: MALLS measurements. Jin Dai: supervision. Priya Subramanian: supervision. Jasper Michels: supervision and editing. Alasdair McKay: investigation. Neil R. Cameron: supervision. Tanja Junkers: conceptualization, writing – reviewing and editing, and supervision.

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Conflicts of interest

There are no conflicts to declare.

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